S4: Contextualised word embeddings

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In this seminar we will look at contextualised word embeddings. Previously we have learned word embeddings and language models as separate representations but which can also be trained in the same end-to-end task of encoding or predicting sequences of words (or characters). In this seminar we will look at the developments of transformers where both tasks have been integrated more closely. (The reasons why trasnformers are called so is because they learn both at the same in order to transform text to another text, or combination of text and other data such as images on either end.)

When predicting a sequence of words, already with LSTMs a mechanism of self-attention has been introduced which weights the importance of the previous generated words for the prediction of the next word. Such represnetations can be stacked on the top of each other and finer representations can be learned. Organising attention heads over several layers allows us to learn structural dependencies between words at different levels of abstraction.

The development lead to pre-training very large language models that capture fine grained syntactic, semantic and world (common-sense) knowledge and can be re-used or fine-tuned for particular NLP tasks whose performance has been greatly improved. However, training and using large-scale models comes at high cost. Firstly, training requires large (and expensive) computing infrastructure (not available to everyone) which consumes a lot of energy. Secondly, the models are trained on texts found on the web, but such texts do not represent all contexts of human activity and social groups (equally or at all) and therefore the resulting models frequently capture particular social biases.

- J. Devlin, M. Chang, K. Lee, and K. Toutanova. BERT: pre-training of deep bidirectional transformers for language understanding. arXiv, arXiv:1810.04805 [cs.CL]:1–14, 2018.
- E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell. On the dangers of stochastic parrots: Can language models be too big? In Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency, FAccT '21, pages 610–623, New York, NY, USA, 2021. Association for Computing Machinery.

Optional further reading (quite advanced):

- A.Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. In Advances in Neural Information Processing Systems, pages 5998–6008, 2017.
- M. Peters, M. Neumann, M. Iyyer, M. Gardner, C. Clark, K. Lee, and L. Zettlemoyer. Deep contextualized word representations. In Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers), pages 2227–2237, New Orleans, Louisiana, June 2018. Association for Computational Linguistics.

- A. Radford, K. Narasimhan, T. Salimans, and I. Sutskever. Improving language understanding by generative pre-training. Technical report, OpenAI, 2018.

Below are some points that you may want to consider when reading these papers.
For the first paper:

- How does BERT compare to (i) word embeddings and (ii) a recursive language model?
- What is the difference between BERT and GPT?
- Details about word tokenisation?
- What is the difference between pre-training and fine-tuning? What is transfer learning?
- Why is BERT trained on masked token and next sentence prediction?
- We hope to have learned syntactic and semantic structure of language and we can extract embeddings for difefferent tokens:
  - What embeddings and and representations at what layers and representations should we use?
  - How does this relate to different tasks?
- What is the role of benchmark datasets such as GLUE and SuperGLUE (a Swedish version called SuperLim)?
  - To what extend and in what ways do these benchmarks evaluate models? What do they evaluate: whether the system can adequately perform a task vs showing that the system is capable of general inference?
  - Have a look at the datasets and consider some examples they contain.

For the second paper:

- Do you agree with the authors of the paper?
- How can we do better NLP?

The paper is quite long and I don't expect that you read it at a fine level. The important thing is that you get familiar with the main arguments it presents. These are also commonly addressed in the current popular science discussions.